

Mechanical Properties of Fluid

Fluid

Fluid is the name of a substance which has the ability to follow when some external force is applied on it.

The study of fluids can be divided into two categories:

1. **Fluid statics:** The branch of physics which deals with the study of fluids at rest is known as fluid statics or hydrostatics.

2. **Fluid dynamics:** The branch of physics which deals with the study of fluid in motion is known as fluid dynamics or hydrodynamics.

Ideal fluid $\rightarrow$ ① Incompressible

Incompressible $\rightarrow$ [Fix volume, mass fix density]

Non-viscous $\rightarrow$ No friction force

Fluid

Liquid & gas

Pressure:-

The normal force acting per unit area of a surface is known as pressure.

If F is the normal force acting on surface is area A then average pressure,

Scalar

$$P = \frac{F}{A}$$

Unit $\Rightarrow$ N/m^2 or Pascal

Other unit $\Rightarrow$ Atmospheric pressure (atm)

$$= 1.013 \times 10^5 N/m^2$$

$$1 \text{ bar} = 10^5 N/m^2$$

$$1 \text{ Torr} = 133 \text{ pa or } N/m^2$$

Fluid pressure

The normal force (or thrust) exerted by a fluid at rest per unit area of the surface placed in contact with it is known as fluid pressure.

Let F = Magnitude of the normal force (thrust) Exerted by a fluid (or liquid) on a surface of area A , placed in contact with the liquid then pressure exerted by the liquid on that surface can be written as

$$P = \frac{F}{A}$$

Some other conventional units of pressure are atmosphere (atm), bar and torr.

$$1 \text{ atm} = 1.013 \times 10^5 \text{ pascal}$$

$$1 \text{ bar} = 10^5 \text{ pascal}$$

$$1 \text{ atm} = 760 \text{ torr} = 1.013 \text{ bar}$$

$$1 \text{ torr} = 133 \text{ pascal}$$



Pressure due to a fluid column

Let the density of any fluid is ρ , there is a point O at a depth h from the surface of the fluid where fluid pressure is to be calculated. O is assumed as center draw a horizontal circle of Area (A)

$$\text{Volume of fluid in cylinder } V = \pi r^2 h$$

$$V = Ah$$

$$\text{mass of fluid in cylinder} = \text{Volume} \times \text{density}$$

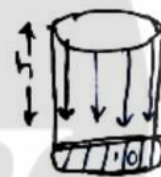
$$m = Ah \times \rho$$

$$\text{Weight of fluid in cylinder } W = mg$$

$$W = Ah \cdot \rho \cdot g \quad \text{--- ①}$$

The force acting on bottom of cylinder = weight of fluid

$$F = Ah \rho g$$



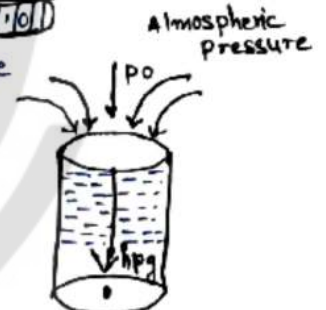
Rest

By definition of fluid pressure

$$P = \frac{F}{A}$$

$$P = \frac{Ah \rho g}{A}$$

$$P = h \rho g$$



$$P = P_0 + h \rho g$$

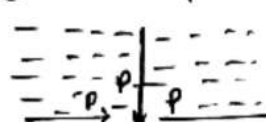
Atmospheric pressure -

The atmospheric pressure at a point is equal to the weight of the air column of atmosphere of unit cross-section extending from that point to top of the atmosphere.

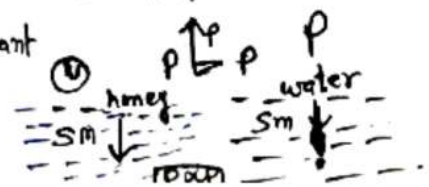
Law of liquid pressure -



(ii) Rest

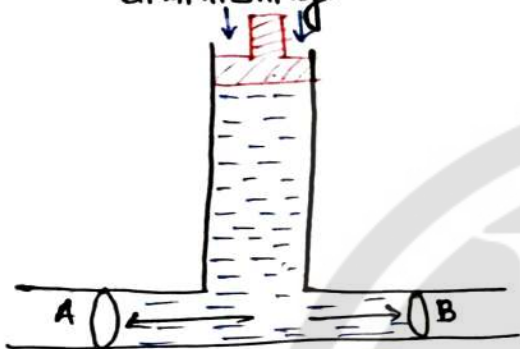


$$P \propto h \quad P = \text{constant}$$



Pascal's Law

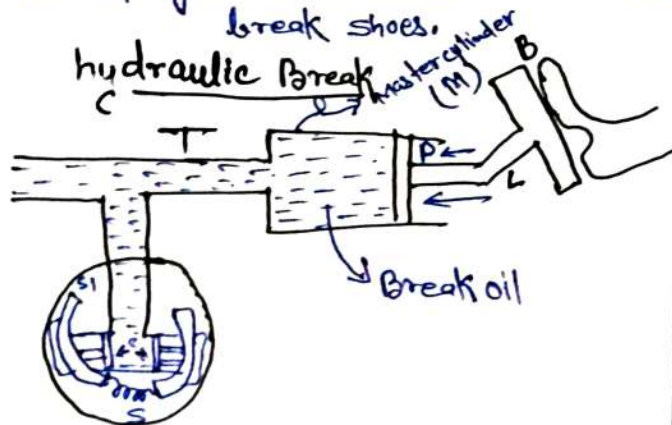
in 1653 Scientist Blaise Pascal proposed a law regarding the transmission of pressure in fluid. This law is known as Pascal's Law or law of pressure transmission in the fluid. According to this Law, if a pressure is applied on any part of the fluid placed in equilibrium inside a vessel then this pressure is transmitted to each part of the fluid in all direction without diminishing.



Hydraulic Break

Principle → it is based on the principle of Pascal's Law.

Construction → it consists of master cylinder (M) which is filled with break oil and provided with air tight piston (P). The piston is connected to the break pedal (B) through lever system (L). Master cylinder is connected to wheel cylinder through tube (T). The wheel cylinder (C) is connected to two pistons (P_1) & (P_2). These pistons are connected to break shoes (S_1) & (S_2) and the spring (S) is connected between two break shoes.



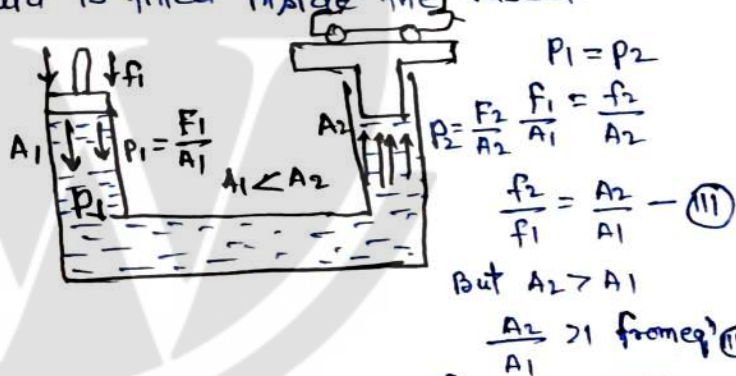
Working → when break pedal (B) is pressed with foot the lever system (L) operates and piston (P) of the master cylinder (M) is pushed inwards. This piston exerts pressure on liquid. According to Pascal's Law, this pressure is transmitted equal to both pistons (P_1) & (P_2) move away from each other then the break shoes also moved away from each other and get attached to the inner rim of the wheel, which retarded the motion of the wheel, due to friction. When the pressure on the break pedal is released, then break shoes return to their initial position by the spring (S).

Hydraulic Lift

it is used to lift the heavy loads

Principle → it is based on the principle of Pascal's Law.

Construction → This is made by connecting two cylindrical vessels of different cross-section area to the end of the horizontal pipe. Each cylindrical vessel contains air tight frictionless pistons and incompressible liquid is filled inside the vessel.



Working → Let the normal force F_1 is applied on the cross-section area, A_1 , of smaller piston, then the pressure applied by piston the pressure acting on liquid by piston

the pressure acting on second piston by liquid

$$P_1 = \frac{F_1}{A_1} \rightarrow \text{--- (1)}$$

$$P_2 = \frac{F_2}{A_2} \rightarrow \text{--- (2)}$$

(According to Pascal's Law) →

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\frac{F_1}{F_2} = \frac{A_1}{A_2} \quad \text{--- (3)}$$

$$A_1 < A_2 \rightarrow \frac{A_1}{A_2} < 1$$

$$\frac{F_1}{F_2} < 1$$

$$F_1 < F_2$$

Density -

Density of a homogeneous fluid is equal to mass per unit volume.

Therefore density of a fluid of mass m having volume v is

$$\rho = \frac{m}{V} \text{ unit} \Rightarrow \text{kg/m}^3$$

$$\frac{[M]}{[L^3]} = [ML^{-3}]$$

(Important points) =

(1) SI unit of density is kg/m^3 and CGS units is gm/cm^3

(2) Dimensional formula of density is $[ML^{-3}]$ This is a scalar quantity.

(3) Density of fluids depends on their pressures and temperatures liquid is incompressible therefore density of a liquid almost remains constant at each pressure. Opposite to this gases show a sufficient change in their densities due to change in its pressure.

(5) if two fluids of densities ρ_1 and ρ_2 and masses m_1 and m_2 are mixed then density of this mixture will be:

$$\rho_{mix} = \frac{\text{Total mass}}{\text{Total volume}} = \frac{m_1 + m_2}{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}}$$

If $m_1 = m_2$, then

$$\rho_{mix} = \frac{2\rho_1 \rho_2}{\rho_1 + \rho_2}$$

$$\rho_{mix} = \frac{m_1 + m_2}{V_1 + V_2} \quad \left[\rho = \frac{m}{V} \right]$$

$$\rho_{mix} = \frac{m_1 + m_2}{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}}$$

$$m_1 = m_2 = m$$

$$\rho_{mix} = \frac{2m}{m \left[\frac{1}{\rho_1} + \frac{1}{\rho_2} \right]} = \frac{2\rho_1 \rho_2}{\rho_1 + \rho_2}$$

(6) if a liquid fluid of density ρ_1 and volume V_1 is mixed with another fluid of density ρ_2 and volume V_2 then density of mixed fluid will be.

$$\rho_{mix} = \frac{m_1 + m_2}{V_1 + V_2} = \frac{V_1 \rho_1 + V_2 \rho_2}{V_1 + V_2}$$

If $V_1 = V_2$ then

$$\rho_{mix} = \frac{\rho_1 + \rho_2}{2}$$

$$\rho_{mix} = \frac{m_1 + m_2}{V_1 + V_2} \quad (\rho = \frac{m}{V})$$

$$\rho_{mix} = \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2}$$

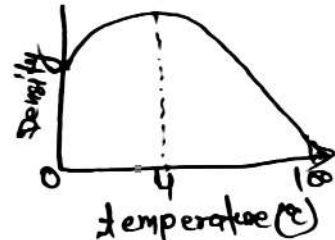
$$V_1 = V_2 = V$$

$$\rho_{mix} = \frac{V(\rho_1 + \rho_2)}{2V}$$

$$\rho_{mix} = \frac{\rho_1 + \rho_2}{2}$$

(4) if two non-mixing fluid of different densities filled in a vessel then fluid of maximum density will be at the bottom and fluid of least density will be at the top

(7) Density of water increases from 0°C to 4°C it is maximum at 4°C and thereafter it decreases from 4° temperature to temperature 100°C (figure)



Formula list

$$\# \rho = \frac{F}{A}$$

Pressure by liquid column

$$p = h\rho g$$

$$\# \text{ Total pressure } p = p_0 + h\rho g$$

Question

The area of each foot of a 50 kg boy is 20 cm^2 , what is the pressure exerted by the boy on the earth, if the (1) stands on one foot, (2) stands on both feet

$$m = 50 \text{ kg}$$

$$A = 20 \text{ cm}^2 = 20 \times 10^{-4} \text{ m}^2$$

$$\text{C1 } p = \frac{F}{A} \quad \text{C2 } p = \frac{F}{2A}$$

$$p = \frac{mg}{A}$$

$$p = \frac{mg}{2A}$$

$$p = \frac{50 \times 10}{20 \times 10^{-4}}$$

$$p = \frac{2.5 \times 10^5}{2}$$

$$p = 2.5 \times 10^5 \text{ N/m}^2$$

$$p = 1.25 \times 10^5 \text{ N/m}^2$$

Question

find the pressure of a swimmer placed at a depth of 10 m from of a lake.

$$\rho = 10^3 \text{ kg/m}^3$$

$$\text{Pressure of liquid column } p = h\rho g$$

$$p = 10 \times 10^3 \times 10$$

$$p = 10^5 \text{ N/m}^2$$

Ideal fluid

Imcompressible
(fixed volume, fixed density, fixed mass)

friction force
No viscous force acting b/w layers of fluid

fluid Dynamics

study
motion of fluid

Stream lined flow -

when a liquid flow such that each particle passing a certain point flow the same path as the preceding particle which passed the same point, such flow is said to be stream lined flow.

Turbulent flow -

when a liquid moves with a high speed the motion of the particle of the fluid becomes irregular. Such a flow is called turbulent flow.

p

Principle of Continuity -

when a fluid flow in stream lined flow through a tube of non-uniform cross-section then at each point in the tube, product of cross-sectional area of the tube and fluid velocity remains constant.

Prove ment -

Let a fluid is following in stream lined flow through a non-uniform cross sectional area of x any y end are A_1 and A_2 . The velocities of the liquid at these ends are v_1 & v_2

(for end)

The distance travelled by liquid in 1 sec = speed $\times$ Time

$$= v_1 \times t$$

$$= v_1$$

volume of entering liquid in

1 sec = area $\times$ dist. travel by liquid in 1 ve

$$= A_1 v_1$$

Mass of entering liquid

in 1 sec = volume of entering liquid in 1 ve $\times$ density

$$= A_1 v_1 \times \rho \quad \text{--- (i)}$$

(for y end)

The distance travelled by liquid in

1 sec = speed $\times$ time

$$= v_2 \times t$$

$$= v_2$$

volume of exiting liquid in

1 sec = Area $\times$ dist travel by liquid in 1 ve

Mass of exiting liquid in 1 sec

= volume of exiting liquid in 1 ve $\times$ density

$$A_2 v_2 \times \rho \quad \text{--- (ii)}$$

from eqn (i) & (ii)

$$A_1 v_1 \times \rho = A_2 v_2 \times \rho$$

$$A_1 v_1 = A_2 v_2$$

$$Av = \text{constant}$$

Note point $\Rightarrow$

$$A_1 v_1 = A_2 v_2$$

$$\pi r_1^2 \cdot v_1 = \pi r_2^2 \cdot v_2$$

$$r_1^2 \cdot v_1 = r_2^2 \cdot v_2$$

$$4 r_1^2 \cdot v_1 = 4 r_2^2 \cdot v_2$$

$$(2 r_1)^2 \cdot v_1 = (2 r_2)^2 \cdot v_2$$

$$D_1^2 v_1 = D_2^2 v_2$$

Bernoulli's theorem

According to this theory, when any liquid flow with stream lined flow in non-uniform cross-section tube, then the sum of pressure energy of fluid per unit volume, kinetic energy of fluid per unit volume and potential energy of fluid per unit volume is always constant.

Bernoulli's Theorem

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$h_1 = h_2$$

Note

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

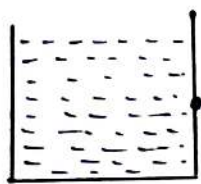
$$\uparrow P_1 - P_2 = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2 \downarrow$$

Note



Velocity of efflux - Torricelli's theorem

When a hole is made in a vessel filled with liquid. So the velocity of the fluid coming out of the hole is equal to that of a body falling from a free plane. This is called Torricelli's Theorem.



$$v = \sqrt{2gh}$$

Acc to third eqⁿ of motion

$$v^2 = u^2 + 2gh$$

$$u = 0$$

$$v^2 = 2gh$$

$$v = \sqrt{2gh}$$

Proof

Let a vessel be filled with a liquid up to height 'H'. Upper end of a liquid is at rest position. There be a hole at depth 'h' below the free surface of the liquid. Hence the liquid is equal and also at the

hole the pressure is equal.

Acc. to Bernoulli's Theorem

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$P_1 = P_2 = P$$

$$\cancel{P} + \frac{1}{2} \rho v_1^2 + \rho g h_1 = \cancel{P} + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$\frac{1}{2} \rho v_1^2 + \rho g h_1 = \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$v_1 = 0, h_1 = H$$

$$v_2 = v, h_2 = (H-h)$$

$$\frac{1}{2} \rho \times 0 + \rho g H = \frac{1}{2} \rho v^2 + \rho g (H-h)$$

$$\rho g h = \frac{1}{2} \rho v^2 + \rho g h - \rho g h$$

$$\frac{1}{2} \rho v^2 = \rho g h$$

$$v^2 = 2gh$$

$$v = \sqrt{2gh}$$

Note Acc to second eqⁿ of motion

$$y = ut + \frac{1}{2} gt^2$$

$$u = 0, y = H-h$$

$$(H-h) = \frac{1}{2} gt^2$$

$$t^2 = \frac{2(H-h)}{g}$$

$$t = \sqrt{\frac{2(H-h)}{g}}$$

Note $R = \sqrt{2gh} \times \sqrt{\frac{2(H-h)}{g}}$

$$R = \sqrt{2gh \times \frac{2(H-h)}{g}}$$

$$R = 2\sqrt{h(H-h)}$$

Venture meter

This instrument is used to measure the rate of flow of liquid in tube.

Rate of fluid flow = Volume of fluid flow in 1 Sec

$$Q = A \times v$$

dist. travel by fluid in 1 Sec = v
Volume of fluid flowing in 1 Sec = Av

$$\begin{aligned} A_1 &> A_2 \\ v_1 &< v_2 \\ P_1 &> P_2 \end{aligned}$$

$$Av = \text{Constant}$$

$$\uparrow A \propto \frac{1}{v} \downarrow$$

Acc to Bernoulli's theorem

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho gh_2$$

$$h_1 = h_2$$

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho gh = P_2 + \frac{1}{2} \rho v_2^2 + \rho gh$$

$$P_1 - P_2 = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2 \quad \text{--- (1)}$$

$$P_1 - P_2 \propto h$$

$$P_1 - P_2 = h \rho g$$

This value putting --- (1)

$$h \rho g = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2$$

$$h \rho g = \frac{1}{2} \rho [v_2^2 - v_1^2]$$

$$2gh = v_2^2 - v_1^2$$

$$2gh = v_1^2 \left[\frac{v_2^2}{v_1^2} - 1 \right]$$

Acc to principle of continuity

$$A_1 v_1 = A_2 v_2$$

$$\frac{v_2}{v_1} = \frac{A_1}{A_2}$$

$$2gh = v_1^2 \left[\frac{A_1^2}{A_2^2} - 1 \right]$$

$$2gh = v_1^2 \left[\frac{A_1^2 - A_2^2}{A_2^2} \right]$$

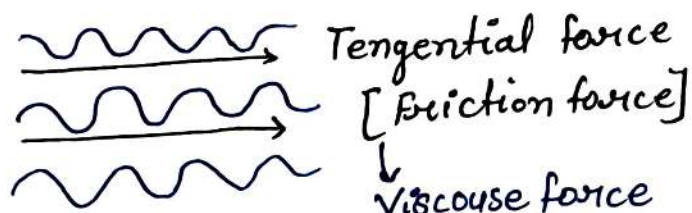
$$v_1^2 = A_2^2 \cdot \frac{2gh}{A_1^2 - A_2^2}$$

$$v_1 = A_2 \sqrt{\frac{2gh}{A_1^2 - A_2^2}}$$

Principle of Venturi meter

$$Q = Av = A_1 v_1$$

$$Q = A_1 A_2 \sqrt{\frac{2gh}{A_1^2 - A_2^2}}$$



$$\text{Velocity gradient} = -\frac{\Delta v}{\Delta n}$$

$$\text{Unit} = \frac{\text{m}}{\text{sec}} \div \frac{\text{m}}{\text{sec}} = \frac{1}{\text{sec}} = \text{sec}^{-1}$$

Viscous force

The viscous force acting between the layers of the fluid depends on the following factors

Viscous force is directly proportional to the velocity gradient

$$F \propto \frac{\Delta v}{\Delta n} \rightarrow 2$$

{from eq(1) & eq(2)}

$$F \propto \frac{\Delta v}{\Delta n} \cdot A$$

$$F = \eta \frac{\Delta v}{\Delta n} \cdot A \quad \text{where } \eta \rightarrow \text{coefficient of viscosity}$$

$$F = \eta A \cdot \frac{\Delta v}{\Delta x}$$

$$\text{If } A = 1 \text{ m}^2$$

$$\frac{\Delta v}{\Delta n} = 1 / \text{sec}$$

$$F = \eta$$

$$f = \eta$$

The co-efficient of viscosity is equal to the viscous force acting per unit area of contact between two layers having unit

velocity gradient between them.

Unit of $\eta \Rightarrow \eta = \frac{f}{A \cdot \frac{\Delta v}{\Delta x}}$

$$\frac{N}{m^2 \cdot 1/sec} = \frac{N \cdot sec}{m^2} = \frac{kg \cdot m \cdot sec}{sec^2 \cdot m^2}$$

$$\frac{kg}{m \cdot sec} \rightarrow m \cdot k \cdot s$$

$$C.G.S \rightarrow \text{Poise}$$
$$1 kg/m \cdot sec = 10 \text{ Poise}$$

Stoke's Law

When a sphere fall down inside the liquid, then viscous force acting in the sphere due to viscosity. It depends upon following points.

Viscous force is directly proportional to radius of sphere
 $F \propto r \rightarrow (1)$

Viscous force is directly proportional to velocity of sphere $\rightarrow (2)$

Viscous force is directly proportional to coefficient of viscosity

$$F \propto \eta$$

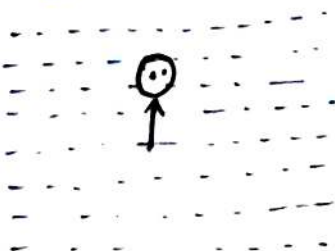
From eq (1), (2) & (3)

$$F \propto r v \eta$$

$$F = 6 \pi \eta v r$$

Lec-5

Upthrust force



upthrust force
 $= \rho \cdot P \cdot g$ density of fluid
Volume of immersed body in fluid

$$\text{Net force } f = f_2 - f_1$$

$$f = P_2 A - P_1 A$$

$$f = A(P_2 - P_1)$$

$$f = Ah P \cdot g$$

$$\text{Upthrust force } f = V \cdot P \cdot g$$

Archimedes principle

Volume of immersed body in fluid
 $=$ Volume of displaced fluid

$$V_s = V_d$$

$$\text{upthrust force} = mg$$

$\rightarrow$ mass of displaced fluid

$$\text{Upthrust force} = \text{weight of displaced fluid}$$

Terminal velocity

The constant velocity attained a force falling body inside the viscous medium is called terminal velocity.

The acceleration of the body moving with terminal velocity is zero.

$$\downarrow \begin{matrix} - f_v + f_u - \\ - \uparrow - \\ - \downarrow - \\ - \downarrow - \\ - \downarrow - \end{matrix}$$
$$w = mg$$

$$f_v + f_u = mg$$

$v = \text{constant}$
[Terminal velocity]

Lec-6

Inter-molecular force

Every substance is made up of molecules. The force of attraction between the molecules is called inter molecular force.



Cohesive force

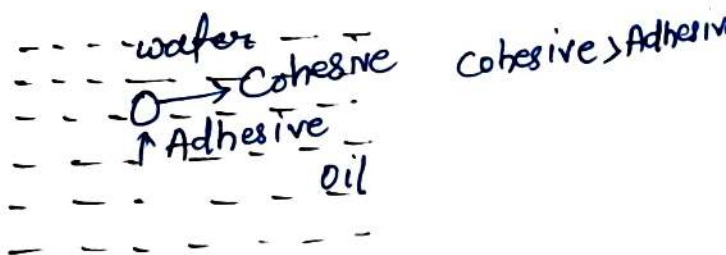
The force of attraction between the molecules of the same substance is called cohesive force.

- force acting between water molecules
- force acting between dust particles.

Adhesive force

The force of attraction between the molecules of the different substance is called adhesive force.

Ex. between water and glass
between oil and water



Surface Tension (S.T)

Ex. small water drop
→ sphere

$$mg = f_v + f_0$$

$$f_v = mg - f_0$$

$$f_v = mg - m'g$$

[m' = weight of displaced fluid]

$$6\pi\eta r \cdot v = (m - m')g$$

$$6\pi\eta r v = (v \cdot \rho - v \cdot \sigma)g$$

$$6\pi\eta r v = v(\rho - \sigma)g$$

$$\cancel{6}\pi\eta r \cdot v = \frac{2}{3}\pi r^3(\rho - \sigma)g$$

$$v = \frac{2}{9} \frac{r^2 g (\rho - \sigma)}{\eta}$$

Formula list

Viscous force $f = \eta A \frac{\Delta v}{\Delta x}$

$f = 6\pi\eta r \cdot v$

Terminal velocity $v = \frac{2}{9} \frac{r^2 g (\rho - \sigma)}{\eta}$

$A_1 v_1 = A_2 v_2$ (Principle of continuity)

$\eta_1^2 v_1 = \eta_2^2 v_2$

$D_1^2 v_1 = D_2^2 v_2$

Bernoulli's Theorem

$$P_1 - P_2 = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2$$

$P = h \rho g$

efflux velocity $v = \sqrt{2gh}$

$$t = \sqrt{2(H-h)/g}$$

$$R = 2\sqrt{h(1t-h)}$$

Ex -

Surface tension is that property of liquid to remain minimum surface are.

$$T = \frac{F_l}{l}$$

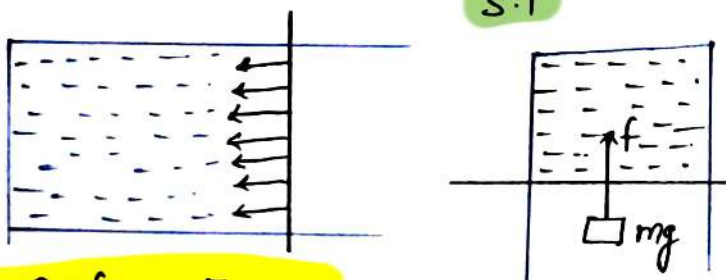
Unit $\Rightarrow$ N/m

Surface Tension

"The surface tension of a liquid is defined as the force per unit length in the plane on a liquid surface acting at right angles on either side of the imaginary line drawn in that surface."

- On mixing a mixture substance in a liquid surface tension of a liquid increase.
- When mixing of soap detergent etc in water then surface tension decreases.
- On increasing the temperature, surface ~~tens~~ tension of a liquid decreases.

Temperature $\propto \frac{1}{S.T}$



Surface Energy -

For increasing the area of the film of any liquid some work

done on it. This work is stored as a potential energy in the film now this potential energy is called surface energy

Surface energy = Work done

$$\Delta U = \Delta W$$

Change in Area $\Delta A = l \times \Delta n$

work done by external agent

$$\Delta W = f \times \Delta n$$

$$\Delta W = T \cdot l \cdot \Delta n \quad \left[T = \frac{f}{l} \right]$$

$$\Delta W = T \times \Delta A$$

By definition of surface energy

$$\Delta U = \Delta W$$

$$\Delta U = T \times \Delta A$$

Excess Pressure

$$F_2 = F_1 + F_e$$

$$F_2 > F_e$$

$$P_1 > P_2 (P_0)$$

Concave) Convex

$$P_{\text{inside}} - P_{\text{outside}} = \frac{2T}{R}$$

$$P_{\text{concave}} - P_{\text{convex}} = \frac{2T}{R}$$

Expression for Excess pressure on drop of water

Change in area $\Delta A = A_2 - A_1$

$$\Delta A = 4\pi R_2^2 - 4\pi R_1^2$$

$$\Delta A = 4\pi [R_2^2 - R_1^2]$$

$$\Delta A = 4\pi [(R + \Delta R)^2 - R^2]$$

$$\Delta A = 4\pi [R^2 + (\Delta R)^2 + 2R \cdot \Delta R - R^2]$$

But $(\Delta R)^2$ is very small.

$$\Delta A = 4\pi \times 2R \cdot \Delta R$$

$$\Delta A = 8\pi R \Delta R$$

By definition of surface energy

$$\Delta U = \Delta W$$

$$T \times \Delta A = F \times \Delta R$$

$$T \times 8\pi R \cdot \Delta R = F \cdot \Delta R$$

$$\frac{F}{A} = \frac{T \times 8\pi R}{A}$$

$$\frac{F}{A} = \frac{T \times 8\pi R}{4\pi R^2}$$

$$P = \frac{2T}{R}$$

Expression for Excess pressure on bubble of water

Let the radius of any bubble is R and its surface tension (T). When a force F acting on this bubble its radius increases by ΔR . Hence the radius of new.

change in area

$$\Delta A = A_2 - A_1$$

$$\Delta A = 4\pi (R_2^2 - R_1^2)$$

$$\Delta A = 4\pi [R + \Delta R]^2 - R^2]$$

$$\Delta A = 4\pi [R^2 + (\Delta R)^2 + 2R \cdot \Delta R - R^2]$$

ΔR^2 is very small

$$\Delta A = 4\pi \times 2R \Delta R = 8\pi R \Delta R$$

But two surface in the bubble

$$\Delta A = 2 \times 8\pi R \Delta R$$

By definition of surface energy

$$\Delta V = \Delta W$$

$$T \times \Delta A = F \times \Delta R$$

$$T \times 16\pi R \cdot \Delta R = F \times \Delta R$$

$$\frac{F}{A} = \frac{T \times 16\pi R}{A}$$

$$\frac{F}{A} = \frac{T \times 16\pi R}{4\pi R^2}$$

$$P = \frac{4T}{R}$$

Surface tension

$$T = \frac{F}{l}$$

Surface energy

$$\Delta U = T \times \Delta A$$

Excess Pressure

Liquid drop

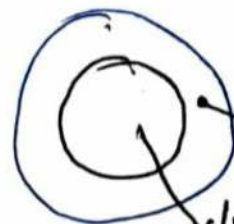


$$P_{in} = P_0 + \frac{2T}{R}$$

Stop Bubble

$$P_{inside} = P_0 + \frac{2T}{R}$$

$$P_{concave} = P_{convex} + \frac{2T}{R}$$



$$P_0 + \frac{4T}{R}$$

Surface tension

$$T = \frac{F}{l}$$

Surface energy

$$\Delta U = T \times \Delta A$$

Excess Pressure

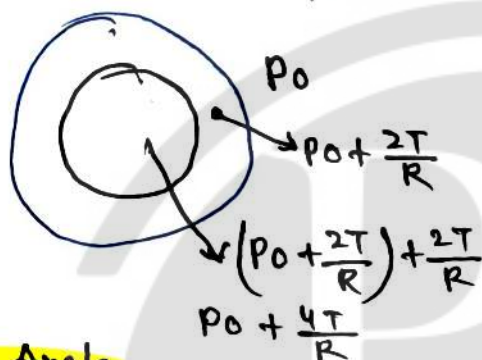
Liquid drop



$$P_{in} = P_{out} = P_0 - \frac{2T}{R}$$

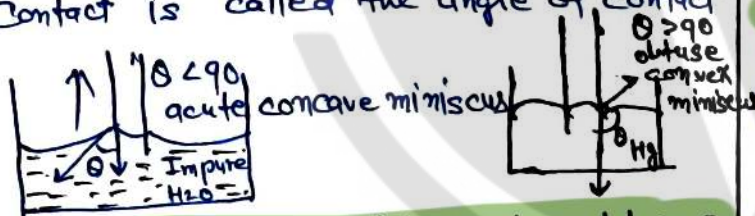
$$P_{inside} = P_0 + \frac{2T}{R}$$

$$P_{concave} = P_{convex} + \frac{2T}{R}$$



Contact Angle

The angle inside the liquid between the tangent to the solid surface and tangent to the liquid surface at the point of contact is called the angle of contact.



Angle of contact b/w water and glass = 0

Angle of contact b/w Hg and glass = 135°

Capillary

When capillary tube is dipped in a liquid then in a capillary tube a curve is drawn due to this pressure in the water is decreases or increases by $\frac{2T}{R}$. For balance the pressure liquid move to up ward or downward in a tube. This Phenomenon is called capillarity.

$$P_y = P_0 \text{ --- (1)}$$

$$P_y = \left(P_0 - \frac{2T}{R}\right) + h\rho g$$

$$P_0 = P_0 - \frac{2T}{R} + h\rho g$$

$$\frac{2T}{R} = h\rho g$$

$$\frac{2T}{R} = h\rho g$$

$$h = \frac{2T}{R\rho g} \text{ --- (2)}$$

In $\triangle OCA$

$$\cos \theta = \frac{r}{R}$$

$$R = \frac{r}{\cos \theta}$$

This value putting in eqn (2)

$$h = \frac{2T}{\frac{r}{\cos \theta} \rho g}$$

$$h = \frac{2T \cos \theta}{r\rho g}$$

